

Reading Guide

what you are seeing on the cover, and how to use it

What is this? A live, interactive 3D viewer showing that a figurate number can be counted two ways. Assembled, a solid is a lattice of dots, and its closed form counts every dot at once. Dissected, it comes apart into tetrahedral atoms, and the Kim identity counts the dots atom by atom. The rail moves between the two, so you can watch one count become the other.

WHAT IS SHOWN ON THE COVER

The page at 1600 by 1000 pixels: the **cube**, $n = 4$, in **Kim Mode**, 55% of the way to Dissect, with the term $4T_{n-1}$ kept. The closed form counts $4^3 = 64$ dots at once. The Kim identity counts them atom by atom: one atom counts $T_4 = 20$ dots, four count $T_3 = 10$ each, and one counts $T_2 = 4$, so $20 + 4 \cdot 10 + 4 = 64$. The kept term lights the 40 dots its four atoms count, each in its atom's color, and dims the rest.

EVERY FEATURE, IN ORDER

- 1

PILLAR II

Opens the guide, with a picture of each part of the page. A first visit gets eight short tips beside the controls instead; **Replay tips**, in the guide, plays them again.
- 2

DUAL

The gold word in the title. Tap it for a short note on the two counts, written out for the solid on screen.
- 3

SHORTCUTS

Space plays or pauses, $\leftarrow$ and $\rightarrow$ move the rail 1% (10% with Shift), **Home** and **End** assemble and dissect, **1**, **2** and **3** choose the view, **A** turns the figure and **R** resets it.
- 4

CLOSED FORM

The left side of the box counts every dot of the assembled solid at once: for the cube, $C_n = n^3$.
- 5

KIM IDENTITY

The right side counts the same dots atom by atom, a tetrahedral number for each atom. Point at a term, or tap it, to light the dots it counts; tap it again to let go.
- 6

THE SIZE, n

Sets the size; the cube runs from 2 to 6. The worked line, the key's counts and the figure follow.
- 7

WORKED LINE

Both counts in numbers:
 $C_4 = 20 + 4 \cdot 10 + 4 = 64$. Its numbers work like the identity's terms.
- 8

THE KEY

Counts the dots, the layers, and the bonds between and within the layers. Point at a row, or tap it, to show that part. Keep the layers, and the figure builds itself layer by layer, each one counted.
- 9

THREE VIEWS

Lattice shows the dots and their bonds. **Kim Mode** lays the atoms over them, and **Kim Mode Only** shows the atoms alone, with a dashed gold outline where the solid stood.
- 10

KIM IDENTITY NOTES

Opens where the atoms come from: Theorem 4.1 of Kim and Lee (2012), why this solid's identity has its terms, and a table of differences. The four small atoms are a reference, the same for every solid.
- 11

THE SEQUENCE

Names the numbers on screen, here the cubic numbers, and links to their entry in the On-Line Encyclopedia of Integer Sequences.
- 12

SIX SOLIDS

Tetrahedral, square pyramidal, octahedral, cubic, stella octangula and centered cube, each tab drawn as its solid. Every control keeps its place as you switch.
- 13

THE WINDOW

Drag the figure to turn it, pinch or scroll to zoom, and move it with two fingers or the right mouse button.
- 14

AUTO-ROTATE

The round arrow turns the figure slowly until you tap it again. Key: **A**.
- 15

RESET VIEW

The crosshair brings the figure back to its starting angle, size and place. Key: **R**.
- 16

PLAY

Runs the rail once toward its far end; a full sweep takes six seconds. Tap it again to pause. Key: **Space**.
- 17

ASSEMBLE

The left end of the rail: tap it to bring the atoms back together, and again while they move to pause them. Key: **Home**.
- 18

THE RAIL

Drag the slider to pull the atoms apart or put them back, to any point between. From Lattice, it opens Kim Mode first.
- 19

DISSECT

The right end: tap it to pull the atoms all the way apart, and again while they move to pause them. The percentage says how far apart they are. Key: **End**.
- 20

AUTO-MORPH

Runs the rail back and forth until you switch it off or take the slider.
- 21

FOOTER

The copyright and the author, his lab, the pillar, the version, the license and the build: where every copy of the page came from.
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ONE LAST THING

If you read this far: choose the stella octangula and drag it until you look straight down on its top point. Its outline becomes the six-pointed star drawn on its tab. There is a lot more to find.

THE SIX SOLIDS, COUNTED TWO WAYS

SOLID	CLOSED FORM	KIM IDENTITY	ATOMS	OEIS
<i>Tetrahedral</i>	$T_n = n(n+1)(n+2)/6$	T_n	1	A000292
<i>Square pyramidal</i>	$P_n = n(n+1)(2n+1)/6$	$T_n + T_{n-1}$	2	A000330
<i>Octahedral</i>	$O_n = n(2n^2+1)/3$	$T_n + 2T_{n-1} + T_{n-2}$	4	A005900
<i>Cubic</i>	$C_n = n^3$	$T_n + 4T_{n-1} + T_{n-2}$	6	A000578
<i>Stella octangula</i>	$SO_n = n(2n^2-1)$	$2T_{2n-1} - O_n$	12	A007588
<i>Centered cube</i>	$CC_n = (n+1)^3 + n^3$	$C_{n+1} + C_n$	12	A005898

$T_n = n(n+1)(n+2)/6$ is the n -th tetrahedral number. Each atom counts one tetrahedral number, so written atom by atom, a count has one term per atom: the stella octangula is $T_n + 10T_{n-1} + T_{n-2}$, and the centered cube $T_{n+1} + 5T_n + 5T_{n-1} + T_{n-2}$. The third difference of a tetrahedral number is 1, so the number of atoms is the third difference of the sequence. Atoms: Theorem 4.1 of H. K. Kim and J. Y. Lee, "Polytope numbers and their properties," arXiv:1206.0511 (2012). Sequences: the On-Line Encyclopedia of Integer Sequences, oeis.org. In the pictures of the page, its 3D window is drawn by a reference renderer that follows three.js r128.